46-887 · Fall 2026

Machine Learning for Business Applications

Week 2: Where do labels come from?

Today's plan

1. What a label is
2. Where labels come from

The basics

What is a label?

A label is a category or value assigned to a piece of data.

A PHOTO	AN EMAIL	A PRODUCT REVIEW	A HOUSE
cat	spam	negative	$615,000
			

Someone or something decides each label.

Kinds of labels

Labels come in different types

TYPE OF LABEL	EXAMPLE QUESTION	POSSIBLE LABELS
Yes or no	Is this transaction fraud?	fraud, legitimate
One of several	What is this support ticket about?	billing, shipping, bug, other
Several at once	Which tags fit this article?	finance, europe, policy
A number	What will this house sell for?	615,000

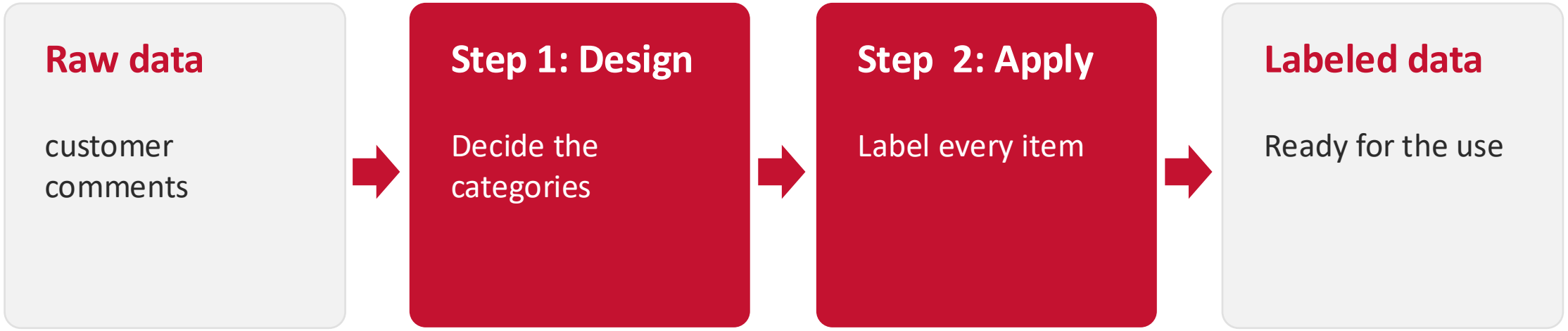
Where labels come from

Four sources of labels

SOURCE	EXAMPLE	STRENGTH	THE CATCH
Customer behavior	clicked, bought, returned	free and automatic	arrives late; may not measure what you care about
People	staff, experts, crowd workers	can handle any question	slow and costs money
Rules	text contains "refund"	fast and consistent	only catches what you thought of
LLMs	prompt a model to label	fast, cheap, flexible	needs clear instructions and and they can be wrong

How it works

Two steps in labeling



Designing the categories and applying them are two separate tasks.

Breakout task

Create the labels

THE SCENARIO

A pizza delivery company

You are the data team.

We collected 20 customer comments.

Label them so we can analyze them later.

1. In a breakout room (18 minutes)
2. Open the file of 20 comments
3. Create labels: one label per comment
4. Write down the categories you used

I won't tell you how. You should decide that as a group.

Breakout task

One more question

IN YOUR GROUP

A manager's question

A manager wants to read the important comments today.

Which ones would you pick?

Does your set of categories make them stand out?

Submit on Canvas

1. Names of your group members
2. The categories you used
3. The comment IDs a manager should read today

Breakout rooms

Discussion

What did you create?

1. **Your categories**
   What are they called?

2. **Why you chose them**
   What did you want the labels to do?

3. **The hardest comment**
   Which one was it, and why?

Good practice

What a label scheme needs

PART	WHAT IT IS	PIZZA EXAMPLE
Categories	the allowed labels	delivery, food quality, price, service, other
Definitions	what each one means	delivery = timing, driver, arrival
Examples	sample comments for each	"Arrived an hour late" is delivery
Edge-case rules	what to do if two fit	praise + late: choose the bigger complaint
An "other" bucket	room for the unexpected	app bugs, support chat

Written down, a scheme works for any labeler: a person or a model.

Chat exercise

Hard cases need rules

COMMENT	WHY IT IS HARD	YOUR RULE
#4 "Amazing pizza, an hour late"	praise and complaint together	?
#10 "Great, another cold pizza."	sarcasm: positive words, negative meaning	?
#12 "Do you have gluten-free options?"	a question, not feedback	?
#9 "Fine."	too short to tell	?
#8 "The app crashed when I paid"	not about the pizza at all	?

In the chat: one sentence that tells a labeler what to do with comment #4.

Cost of mistakes

What happens when a label is wrong

MISTAKE 1

An ordinary comment, mislabeled

"Arrived an hour late" is filed under food quality instead of delivery.

One bar in a chart is slightly off.

Small.

MISTAKE 2

A safety comment, missed

"The dairy-free pizza had cheese on it and I felt sick all night" is filed as an ordinary negative comment.

Nobody reads it today.

A sick customer, possible legal exposure.

Two different mistakes with very different costs. Accuracy treats them the same.

Cost of mistakes

Two ways to be wrong

	WHAT WENT WRONG	WHAT IT COSTS	WHO PAYS
False alarm	flagged as urgent, but it is not	a manager spends a minute reading it	the manager's time
Missed case	urgent, but not flagged	a harmed or lost customer, a legal problem	the customer, then the company

Which mistake would you rather make?

The cost of each mistake is a business input, decided before you build.

A prompt

Your sentence

from the chat exercise

Instructions

for the labeler

A person or an LLM

applies them

```
You label customer comments for a pizza delivery company.
Categories: delivery, food quality, price, service, other.
Rule: if a comment praises the food but complains about
      timing, label it "delivery".
Return only the category name.
```

A prompt is a labeling guide for a machine.

Measuring labels

We built a model to label customer comments. It is 100% accurate. Should we trust it?

The big picture

Four decisions before any model

Who labels?

A person, an expert, the customer, or an LLM?

What are the categories?

They follow from the business question.

What does a mistake cost?

Which errors are cheap, and which are not?

How do we handle disagreement?

Measure it, resolve it, or accept it?

These are business decisions, made before the model exists.

LAST QUESTION

Think of one business decision where the choice of categories would change what someone does.

Type it in the Zoom chat. Do not press send yet.

Who chose the categories? What changes if they are different?